AZURESKY PTY LTD PROFICIENCY TESTING

GENERAL ANALYSIS REPORT

Laboratory Code: 67

Final Report

Report No: 16

Date Issued: 04th March 2026

Document Compiled by:

Laboratory Supervisor – Tshepo Maponya

Signature: 

Document Authorised by:

Managing Director - Nonhlanhla Msibi

Signature: 

1. Letter to participant

Azuresky (Pty) Ltd would like to express gratitude to **Azuresky Pty Ltd** for participating in this round of the proficiency testing scheme.

- Each participant is allocated a random, unique number for the programme to enable total confidentiality of results.
- The unique identification number for your laboratory is **67.**

Participants are encouraged to submit any comments or queries regarding the results or operation of the scheme to info@azuresky.it.com or nonhlanhla@azuresky.it.com.

2. Executive summary

Azuresky Pty Ltd is a proficiency testing scheme provider and certified reference material producer. We aim to produce a homogeneous proficiency test sample and stable certified reference material to be used for the evaluation of participant performance against pre-defined criteria in interlaboratory comparison programs specific to coal testing methods and calibration of coal testing equipment.

The source of the coal samples for this round was acquired from a mine located in Mpumalanga, and the coal type is bituminous.

The identity of the PT samples dispatched for this round is “Proficiency Testing February 2026”.

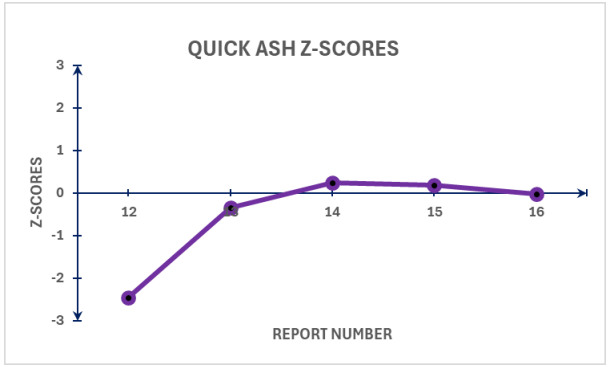
The total number of participants for this round was 27.

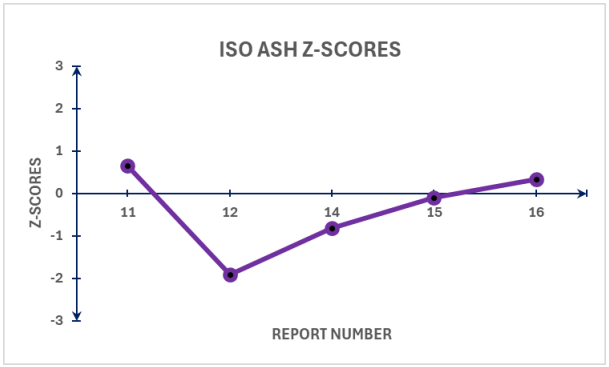
When the sample size is sufficiently large, the sampling distribution of the sample mean approaches a normal distribution, regardless of the shape of the original population distribution. This assumption underpins all statistical analyses and calculations presented in this report.

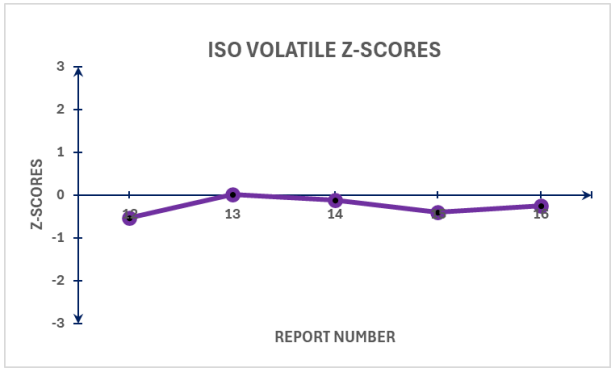
One significant outliers was identified in this report, relating to ISO Volatile Matter.

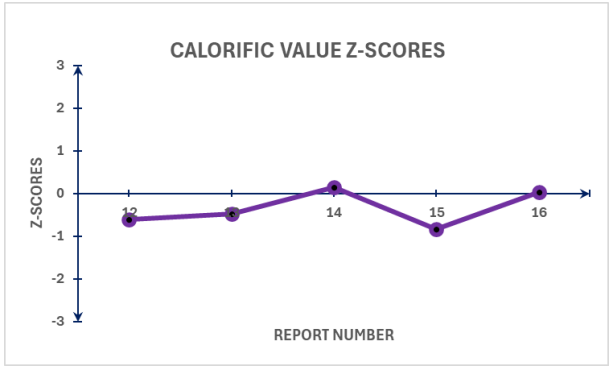
In accordance with SANAS R80, small interlaboratory comparisons are defined as those with seven or fewer participants. As this scheme is not classified as a small ILC, Azuresky applies an internal threshold of n ≥ 10 for the calculation and reporting of z-scores. This criterion is applied to ensure that statistical estimates (mean, standard deviation, and z-scores) are robust and not unduly influenced by small datasets. Where the number of participants is fewer than 10, only indicative results on an air-dry basis are provided.

Interpretation of trends requires a minimum of five data points to ensure a meaningful analysis, while graphical representation necessitates at least two data points to establish a visual comparison.









- **Quick Ash:** Acceptable Trend
- **ISO Ash:** Acceptable Trend
- **ISO Volatile:** Acceptable Trend
- **Calorific Value:** Acceptable Trend

3. List of participants

No	Lab Name
1	Azuresky Pty Ltd
2	Best Enough Laboratory Services NCC
3	Best enough Springs
4	Bestech Zomhlaba Lakeside
5	Eskom Kendal Powerstation
6	Eskom Research and Testing
7	Imbally Coal Laboratory
8	Imbally uMlabu Colliery
9	Mfulawamanzi Kleinfontein
10	Mfulawamanzi Main Laboratory
11	Noko Analytical Services Witbank
12	Sibonisiwe Coal Laboratory Services Arnot
13	Sibonisiwe Coal Laboratory Services Clewer
14	Sibonisiwe Coal Laboratory Services Main Lab
15	Sibonisiwe Coal Laboratory Services RMC
16	Sibonisiwe Coal Laboratory Services Wescoal
17	Siza Coal Services (Pty) Ltd Sasolburg
18	Siza Coal Services Dundee
19	Siza Coal Services Kinross
20	Siza Coal Services Leewpan
21	Siza Coal Services Matsambisa
22	Siza Coal Services Mgayo
23	Siza Coal Services Middelburg
24	Siza Coal Services NBC
25	Umzamo Analytical Services Main lab
26	Umzamo Ananlytical Services Hendrina
27	Vitrovian Analytical Services

4. Homogeneity check

10 random samples were extracted from the batch during packaging. These were used for homogeneity assessment prior to the distribution of samples to participants.

	Dry basis				
# Sample	%R1	%R2	%Sample av (Xt)	Range (Wt)	Range sqd
1	22,66	22,81	22,73	0,1505141	0,0227
2	22,67	22,73	22,70	0,0528541	0,0028
3	22,85	22,77	22,81	0,0772957	0,0060
4	22,85	22,72	22,78	0,1282999	0,0165
5	22,56	22,61	22,58	0,0562482	0,0032
6	22,79	22,89	22,84	0,1007301	0,0101
7	22,74	22,82	22,78	0,0763109	0,0058
8	22,69	23,14	22,91	0,4491311	0,2017
9	23,27	23,15	23,21	0,1283765	0,0165
10	23,18	23,39	23,29	0,209926	0,0441
Overall Average			22,86		
Standard Deviation			0,23		
SSwithin			1,016		
SSbetween			0,0290		
Õbetween			0,7025		
õpt			4,116		
Check value			1,235		

where: õpt is the standard deviation from the pt

Conclusion : Õbetween< Check value. Thus, homogeneity is sufficient.

5. Stability check

10 samples which were used for the homogeneity assessment were analysed to check for stability, the results are as following:

# Sample	%R1	%R2	Sample av (Xt)	Range (Wt)	Range sqd
1	23,11	23,08	23,10	0,03	0,00
2	23,05	22,92	22,99	0,13	0,02
3	23,15	23,11	23,13	0,04	0,00
4	23,22	23,13	23,17	0,08	0,01
5	23,05	23,20	23,13	0,16	0,02
6	23,50	23,48	23,49	0,01	0,00
7	23,28	23,34	23,31	0,06	0,00
8	23,23	23,24	23,24	0,01	0,00
9	23,37	23,25	23,31	0,12	0,01
10	23,35	23,40	23,37	0,05	0,00
Overall Average			23,22		
Standard Deviation			0,15		
SSwithin			0,435		
SSbetween			0,001		

Check value from the homogeneity assessment is	1,234681598
The mean difference between homogeneity and stability assessments	**0,35917845**

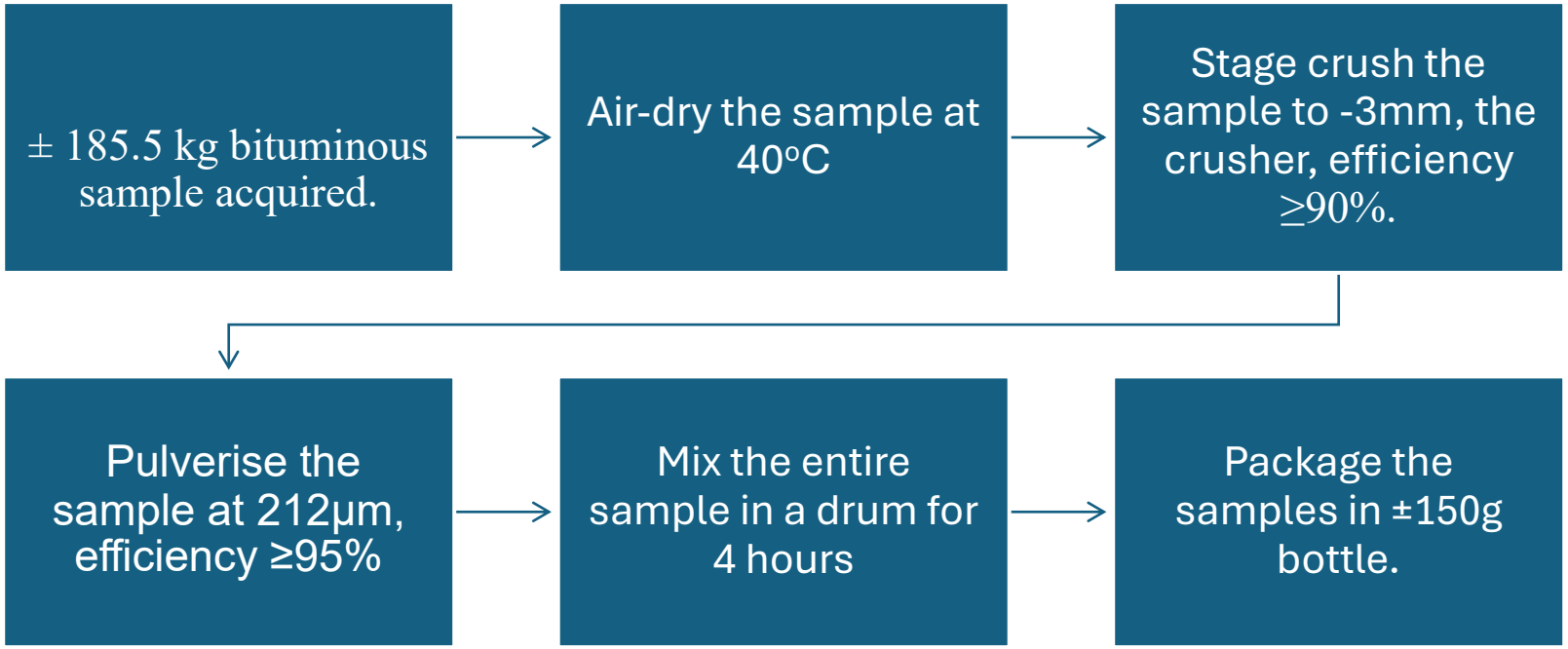
Since 0,36<1,23, the PT samples meet stability requirements

6. Sample preparation

The sample preparation procedure was conducted in accordance with the guidelines outlined in ISO 13909-4, ISO/IEC 17025, ISO/IEC 17043, and ISO 17034. Before the distribution of samples, ten random sub-samples of approximately 150 grams each were drawn using the flat-and-heap method from the 600 grams obtained from the homogeneity drum.

None of the activities for the PT program were subcontracted.

Sample preparation process flow:



7. Measurement Protocol and identification of the measurement standard (ISO/IEC 17043:2023, clause 7.1.2)

The analysis performed for this round, together with the SI units, is stipulated below:

- Calorific Value measured in MJ/Kg
- ISO Ash in % m/m
- Quick Ash in % m/m
- Total Sulphur %
- Moisture in Analysis sample % m/m
- Ash Fusion Temperature (°C)
- Hydrogen %
- Total Carbon %

8. Measurement Results

	AIR DRY BASIS						ASH FUSION TEMPERATURE (oC)					
Lab ID	**%Moisture**	**%Quick Ash**	**% ISO Ash**	**% ISO Volatile**	**Calorific Value MJ/Kg**	**%Total Sulphur**	**DT**	**ST**	**HT**	**FT**	**%Hydrogen**	**%Total Carbon**
1	4,8	22,30			22,59	0,91						
2	5,2	22,10	22,00	25,28	24,28	0,78						
5	4,6	22,10	22,40	23,80	22,47	0,95						
6	5,1	22,40	22,20	23,80	22,37	0,93	1500	1500	1500	1500		
7	5,0	22,00	22,60	23,10	22,51	0,89						
12	4,82		21,70	23,48	22,68	0,96						
15	5,2	21,90	22,20	22,90	23,01	0,9						
17	5,2	21,90	22,10	23,40	22,52	1,02						
19	5,6	22,30	21,90	23,30	22,87	0,90	1500	1500	1500	1500	2,94	30,5
20	4,9		22,50	23,50	21,38	0,83						
25	4,8	22,40	22,10	23,40	22,50	0,99	1490	1500	1500	1500		
27	4,7	21,90		23,60	22,46	0,90						
28	4,8		22,20	23,40	22,55	0,86						
29	3,8	22,50		23,50	22,39	0,88						
40	4,6		22,34	23,41	22,78	0,86						
44	5,2	22,30		23,70	23,50							
46	5,3		22,20	23,20	22,80	0,92						
47	3,4	22,50		23,90	23,51	0,78						
63	5,2	22,10	22,00	28,90	22,44	0,99						
65	4,5	22,30	22,20	23,80	22,83	0,92	1500	1500	1500	1500		
66	4,4	22,20	22,02	25,07	24,52							
67	5,4	22,08	22,19	23,29	22,70							
68	5,0	22,50	22,40	24,30	22,46	0,98						
69	5,1		22,60	23,20	22,53	0,97						
71	4,2	22,40		24,10	22,80	0,88						
72	5,4	21,90		23,60	23,23							
74	4,2	22,40		24,11	23,02							

	Dry-basis										
Lab ID	% ISO Moisture	% Quick Ash	Z-score	% ISO Ash	Z-score	%Volatile	Z-score	Calorific Value (MJ/Kg)	Z-score	%Total Sulphur	Z-score
1	4,8	23,42	0,50	***	***	***	***	23,73	-0,37	0,96	0,01
2	5,2	23,31	-0,12	23,2	-0,61	26,67	1,70	25,61	2,62	0,82	-1,91
5	4,6	23,17	-0,94	23,5	0,46	24,95	0,06	23,55	-0,65	1,00	0,59
6	5,1	23,60	1,50	23,4	0,12	25,08	0,19	23,57	-0,62	0,98	0,36
7	5,0	23,16	-0,98	23,8	1,68	24,32	-0,54	23,69	-0,42	0,94	-0,27
12	4,8	***	***	22,8	-2,22	24,67	-0,20	23,83	-0,21	1,01	0,77
15	5,2	23,10	-1,29	23,4	0,22	24,16	-0,69	24,27	0,50	0,95	-0,08
17	5,2	23,10	-1,29	23,3	-0,20	24,68	-0,19	23,76	-0,32	1,08	1,74
19	5,6	23,62	1,60	23,2	-0,64	24,68	-0,19	24,23	0,42	0,95	-0,03
20	4,9	***	***	23,7	1,17	24,71	-0,16	22,48	-2,35	0,87	-1,19
25	4,8	23,53	1,09	23,2	-0,58	24,58	-0,29	23,63	-0,52	1,04	1,22
27	4,7	22,98	-1,97	***	***	24,76	-0,11	23,57	-0,62	0,94	-0,16
28	4,8	***	***	23,3	-0,17	24,58	-0,29	23,69	-0,43	0,90	-0,75
29	3,8	23,39	0,30	***	***	24,43	-0,43	23,27	-1,09	0,91	-0,58
40	4,6	***	***	23,4	0,22	24,54	-0,33	23,88	-0,13	0,90	-0,77
44	5,2	23,52	1,05	***	***	25,00	0,11	24,79	1,32	***	***
46	5,3	***	***	23,4	0,31	24,50	-0,37	24,08	0,18	0,97	0,24
47	3,4	23,29	-0,23	***	***	24,74	-0,14	24,34	0,60	0,81	-2,13
63	5,2	23,31	-0,11	23,2	-0,60	30,49	5,34	23,67	-0,45	1,04	1,29
65	4,5	23,35	0,09	23,2	-0,46	24,92	0,04	23,91	-0,09	0,96	0,12
66	4,4	23,21	-0,68	23,0	-1,33	26,21	1,27	25,64	2,66	***	***
67	5,4	23,33	-0,02	23,4	0,33	24,61	-0,26	23,99	0,04	***	***
68	5,0	23,68	1,95	23,6	0,85	25,58	0,66	23,64	-0,50	1,03	1,10
69	5,1	***	***	23,8	1,78	24,45	-0,42	23,74	-0,35	1,02	0,97
71	4,2	23,38	0,27	***	***	25,16	0,26	23,80	-0,25	0,92	-0,53
72	5,4	23,15	-1,02	***	***	24,95	0,06	24,56	0,95	***	***
74	4,2	23,39	0,29	***	***	25,17	0,27	24,03	0,12	***	***

# of participants	-	27	21	-	18	-	26	-	27	-	22	-
Outilers	-	0	0	-	0	-	1	-	0	-	0	-
# of particpants after outlier elimination	-	27	21	-	18	-	25	-	27	-	22	-
Average	-	4,8	23,3	-	23,4	-	24,9	-	23,96	-	0,96	-
Standard deviation	-	0,51	0,18	-	0,25	-	1,05	-	0,63	-	0,07	-
%RSD	-	10,47	0,77	-	1,09	-	4,22	-	2,63	-	7,25	-
Median	-	-	23,3	-	23,4	-	24,7	-	23,80	-	0,95	-
UoM	-	-	0,04		0,06		0,21		0,12		0,01	
Min	-	3,4	23,0	-	22,8	-	24,2	-	22,48	-	0,81	-
Max	-	5,6	23,7	-	23,8	-	30,5	-	25,64	-	1,08	-
standard error	-	0,20	0,09	-	0,13	-	0,43	-	0,25	-	0,03	-
Lower confidence limit	-	4,6	23,2	-	23,2	-	24,5	-	23,71	-	0,92	-
Upper confidence limit	-	5,0	23,4	-	23,5	-	25,3	-	24,21	-	0,99	-
Lower confidence limit 3x SE	-	-	23,1	-	23,0	-	23,6	-	23,21	-	0,86	-
Upper confidence limit 3x SE	-	-	23,6	-	23,8	-	26,2	-	24,71	-	1,05	-

9. Reference Value(s) and their establishment (SANAS R80, clause 7.3)

The assignment of reference values has been carried out in accordance with the requirements of ISO/IEC 17043:2023 (clause 7.7) and ISO 17034:2016 (clauses 7.12–7.13). These standards provide internationally accepted guidance on the establishment of reference values, including the traceability of results, appropriate statistical treatment, and the evaluation of measurement uncertainty.

Disclosure of Assigned Values

In line with the requirements of **ISO/IEC 17043:2023, clause 7.2.3.5**, measures are in place to ensure that participants cannot gain an advantage through early disclosure of assigned values. The following practices are applied:

- **Single Submission:** Each participant is allowed to submit their results only once. After submission, no edits or resubmissions are permitted.

- **Post-closure Analysis:** All statistical evaluation and assignment of values are performed only after the submission deadline has passed.
- **Simultaneous Release:** Final PT reports, including assigned values, are issued to all participants at the same time. This ensures that no participant has access to assigned values ahead of others.

Through these practices, the integrity of the proficiency testing process is maintained, and participants are assured of fair and unbiased evaluation.

10. Evaluation of the measurement results

- Outliers
  - ISO Volatile, Lab ID 63 (see dry base table for z-scores)
- All results are accepted as correct and cannot be changed after the release of the report.

Z-Score Evaluation Criteria:

- A z-score within the range of **±1** is considered **acceptable**.
- A z-score within the range of **±2** is regarded as **satisfactory**.
- A z-score between **±2 and ±3** is classified as **questionable**, indicating potential issues that warrant further investigation.
- A z-score exceeding **±3** is deemed **unsatisfactory** and is treated as an **outlier.**

Figure 1: Volatile matter z-scores

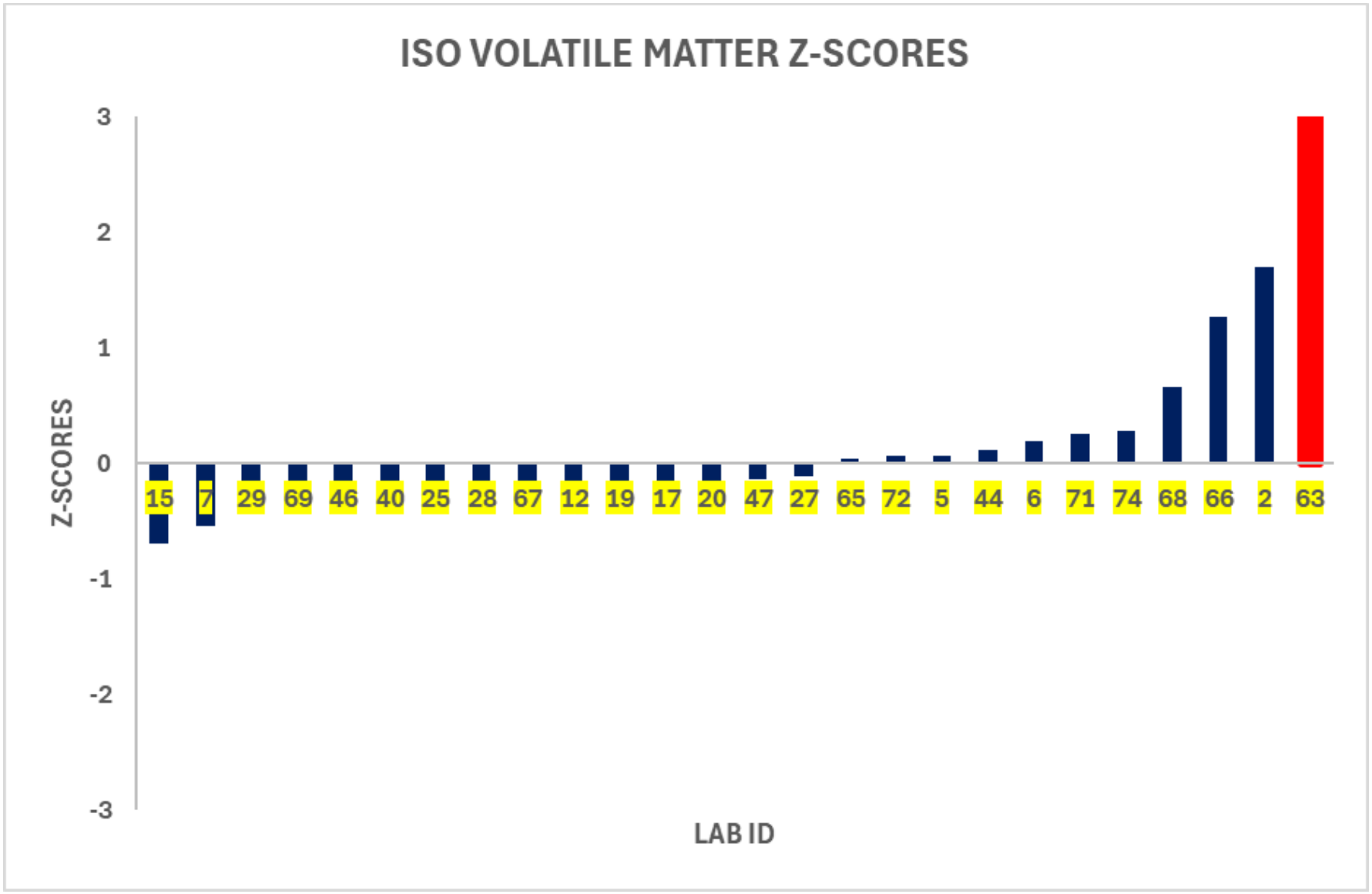


figure 2: ISO ash z-scores

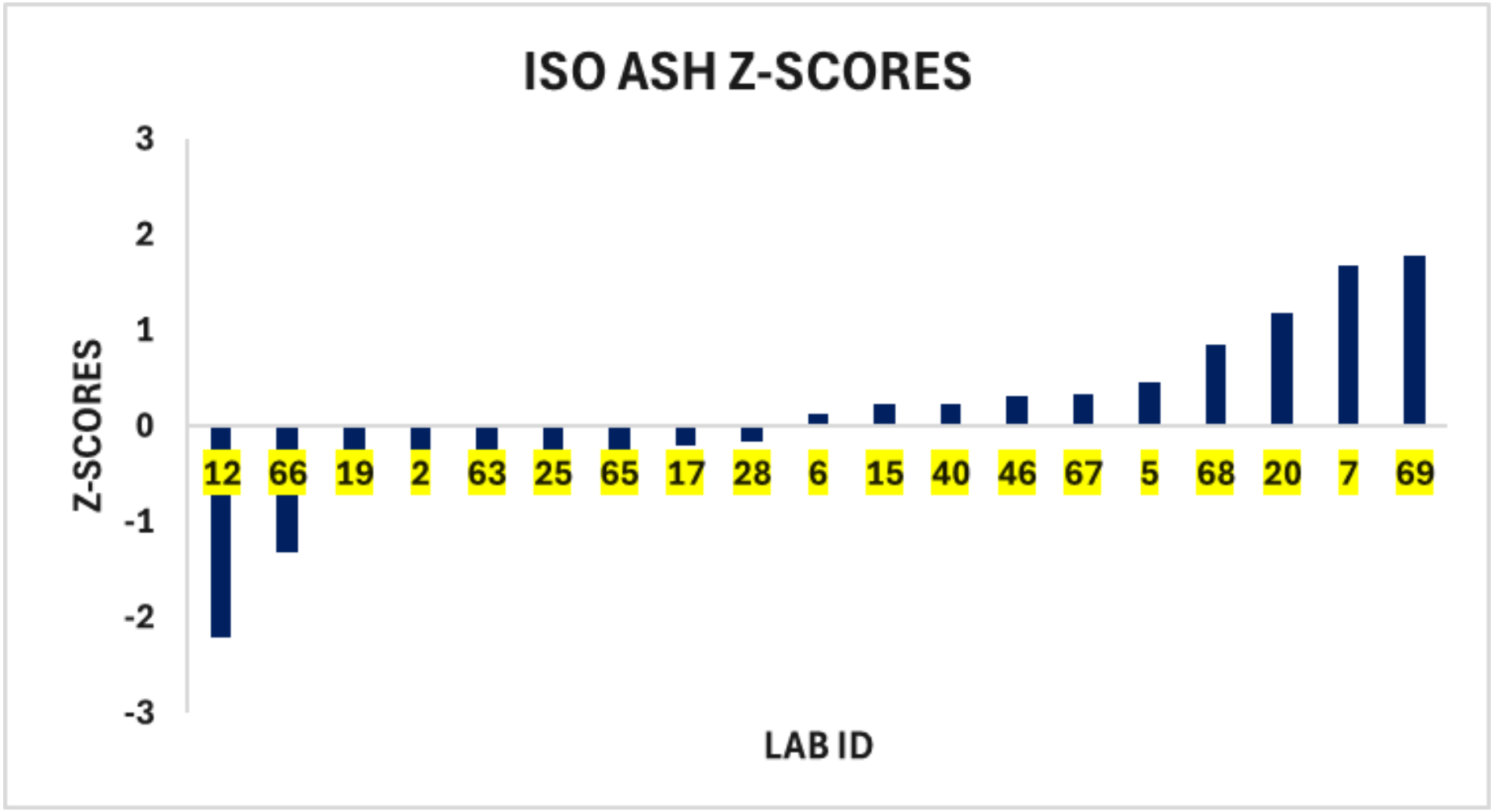


Figure 3: Quick Ash z-scores

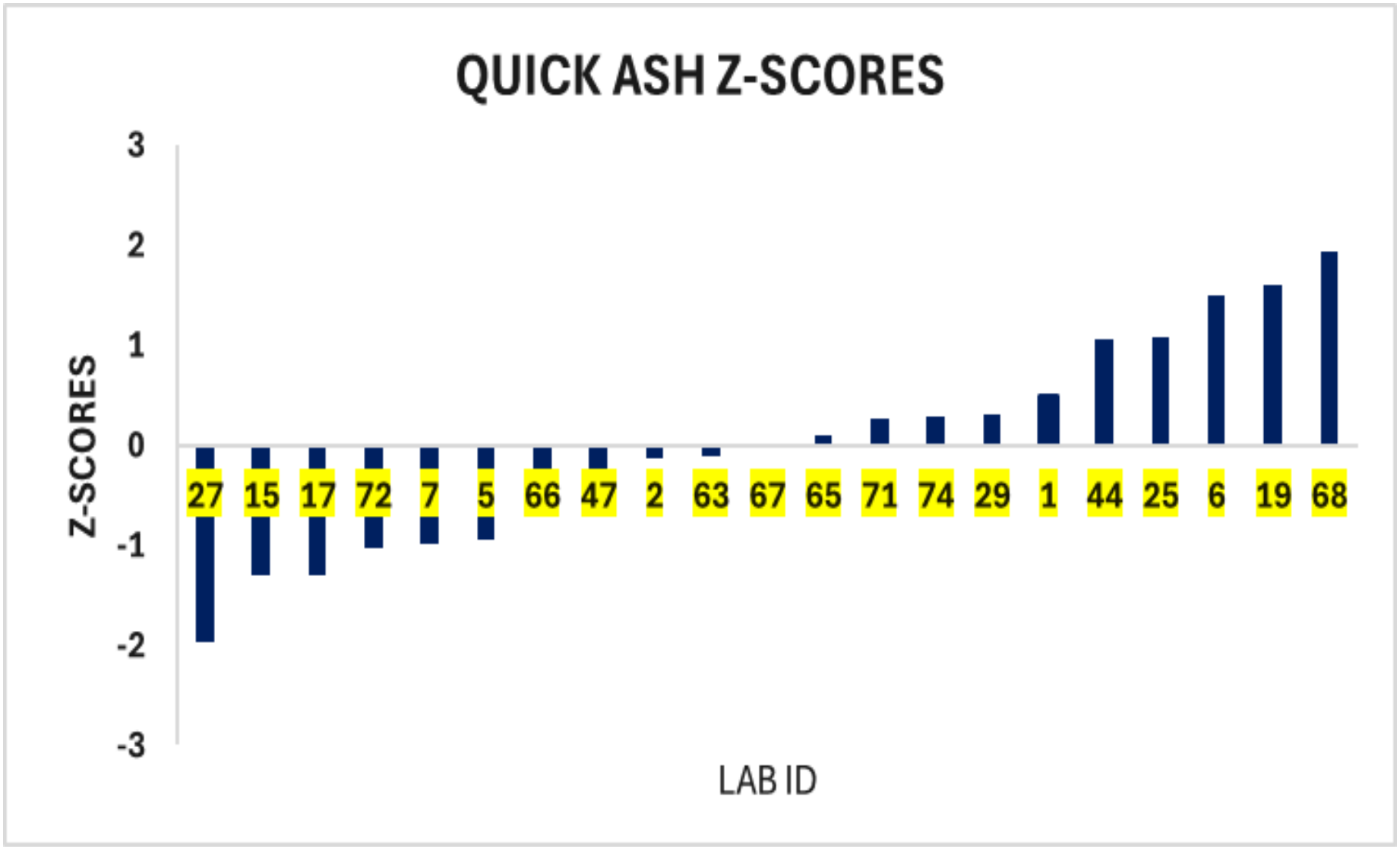


Figure 4: Calorific Value z-scores

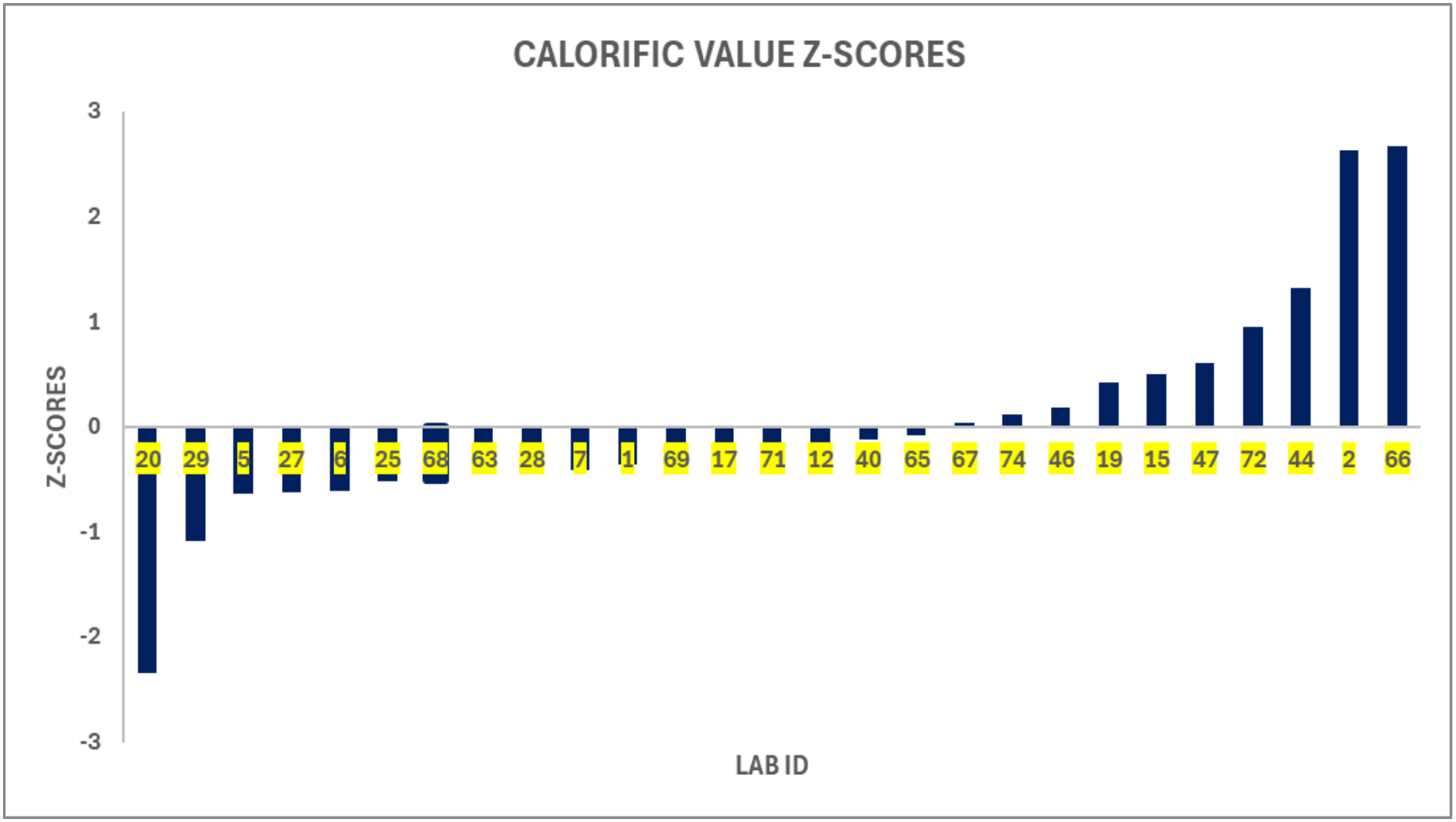
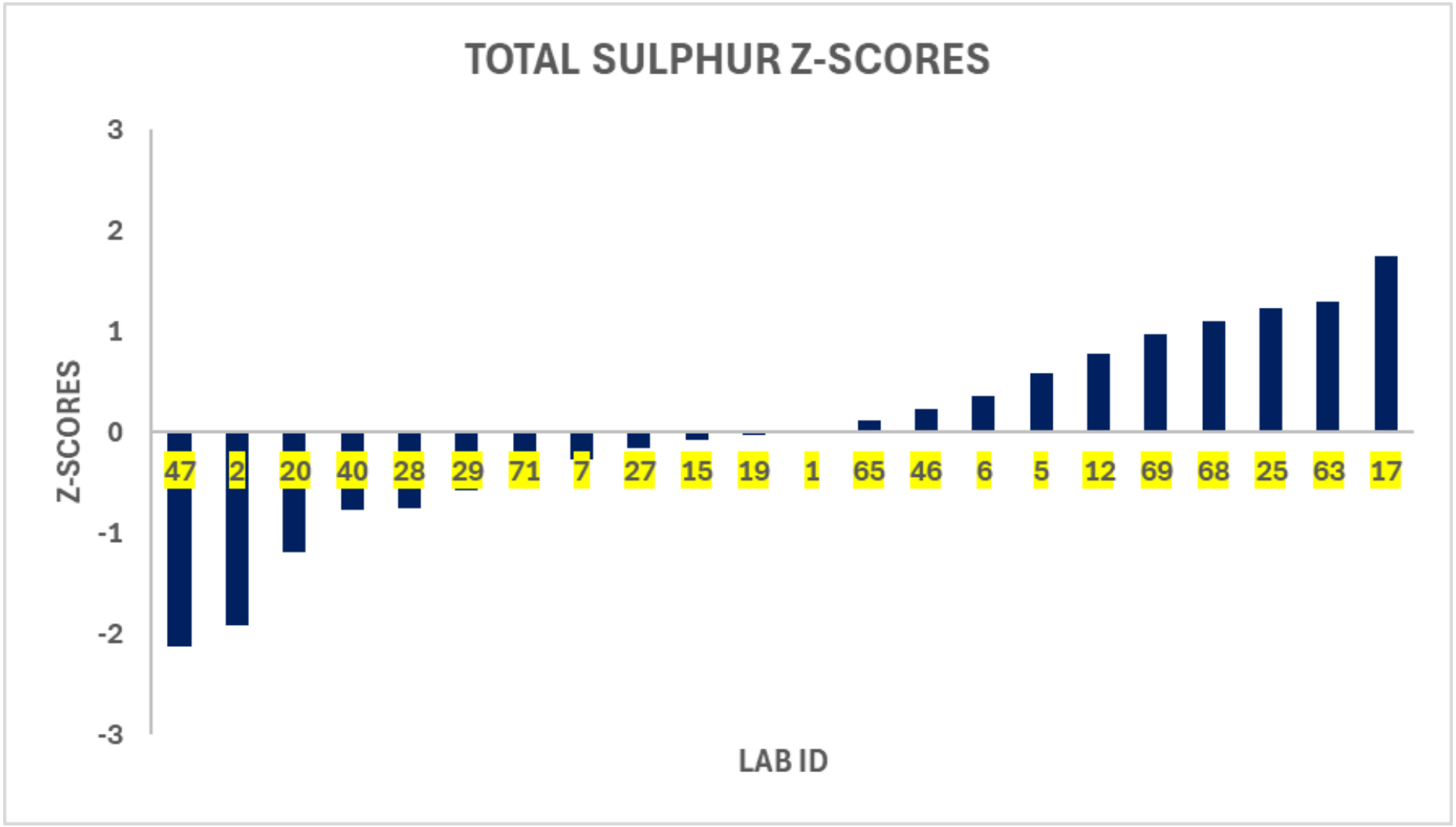


Figure 3: Total Sulphur z-scores



11. Conclusion

11.1 ISO Moisture

- **Results** from **27** laboratories ranged from **3.4%** to **5.6%**, with a mean of **4.8%** and a standard deviation of **0.51%**.
- **No** statistical outliers were identified.
- The calculated **%RSD was 10.47%**, showing moderate to high inter-laboratory variability.
- The **95% confidence interval** for the mean was **4.6% to 5.0%**.
- **Conclusion**: Results demonstrate acceptable performance, though the distribution suggests environmental conditions significantly influenced laboratory outcomes.

11.2 Quick Ash

- **Results** from **21** laboratories ranged from **23.0%** to **23.7%**, with a mean of **23.3%** and a standard deviation of **0.18%**.
- **No** statistical outliers were detected.
- The **%RSD was 0.77%**, reflecting excellent inter-laboratory precision.
- The **95% confidence interval** was **23.2% to 23.4%**.
- **Conclusion**: Quick ash determinations show very high consistency, confirming high method stability across the participant pool.

11.3 ISO Ash

- **Results** from **18** laboratories ranged from **22.8%** to **23.8%**, yielding a mean of **23.4%** with a standard deviation of **0.25%**.
- **No** outliers were identified.
- The **%RSD was 1.09%**, confirming good repeatability and reproducibility.
- The **95% confidence interval** for the mean was **23.2% to 23.5%**.
- **Conclusion**: ISO Ash results show strong alignment and strict adherence to standardized analytical procedures.

11.4 Volatile Matter

- **Results** from **25** laboratories (after outlier removal) ranged from **24.2%** to **26.7%,** with a mean of **24.9%** and a standard deviation of **1.05%**.
- **One** statistical outlier (**Lab ID 63**, Z = 5.34) was identified and excluded.
- The **%RSD was 4.22%**, indicating moderate variability.
- The **95% confidence interval** for the mean was **24.5% to 25.3%**.
- **Conclusion**: Volatile matter results are statistically comparable across laboratories, with data clustering tightly around the consensus value once the outlier was removed.

11.5 Calorific Value

- **Results** from **27** laboratories ranged from **22.48 MJ/kg** to **25.64 MJ/kg**, with a mean of **23.96 MJ/kg** and a standard deviation of **0.63 MJ/kg**.
- **No** outliers were detected.
- The **%RSD was 2.63%**, indicating good inter-laboratory precision.
- The **95% confidence interval** for the mean was **23.71 MJ/kg to 24.21 MJ/kg**.
- **Conclusion**: Calorific value results demonstrate robust performance and consistent calibration across the participating laboratories.

11.6 Total Sulphur

- **Results** from **22** laboratories ranged from **0.81%** to **1.08%**, with a mean of **0.96%** and a standard deviation of **0.07%**.
- **No** outliers were detected.
- The **%RSD was 7.25%**, reflecting the typical variability found in sulphur analysis at these concentrations.
- The **95% confidence interval** for the mean was **0.92%to 0.99%**.
- **Conclusion**: Sulphur determinations show acceptable reproducibility, confirming consistent analytical control across most participants.

Overall Summary

Across all measured parameters for this round, the inter-laboratory results reflect sound precision, consistent method control, and high analytical integrity. Both Quick Ash and ISO Ash determinations demonstrated the highest level of reproducibility, with %RSD values of 0.77% and 1.09% respectively. Calorific Value also showed strong alignment with a %RSD of 2.63% and no identified outliers.

While ISO Moisture and Total Sulphur exhibited higher relative variability (10.47% and 7.25% respectively), these levels are consistent with expectations for these specific parameters at the reported concentration ranges. Only one outlier was detected across all laboratory data points for this sample, indicating a high degree of homogeneity and consensus among participants.